

ARIEF F-SA WIJAYA

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PROFILE SUMMARY

Fourth-semester Diploma (D3) student in Informatics Engineering at Bandung State Polytechnic with a strong interest in software development and structured solution design. Experienced in thoroughly analyzing problems, designing clear solution architectures, and managing complexity prior to implementation. Has a background in competitive programming that fosters a systematic, analytical, and efficiency-oriented mindset, enabling the development of modular, readable, and maintainable code. Prioritizes architectural quality, code readability, and long-term maintainability in every software solution.

TECHNICAL SKILLS

Bahasa Pemrograman

C++, Python, JavaScript, PHP, Java

Web Application Development

- Building well-structured web applications
- Designing frontend–backend data flows using API- and HTTP-based communication
- Managing application architecture using the MVC approach

Software Engineering

- Problem solving and solution design based on requirements analysis
- Writing modular, readable, and maintainable code
- Debugging and refactoring

Algorithms & Data Structures

- Advanced Data Structures, Dynamic Programming, Graph Algorithms, Binary Search
- Time and space complexity analysis

EDUCATION

Politeknik Negeri Bandung (*Bandung State Polytechnic*)

2024-Present

Diploma (D3) Program in Informatics Engineering

SMAN 1 Subang (*Senior High School 1 Subang*)

2021-2024

Math and Natural Science

PROJECT EXPERIENCE

Study Buddy Education

January 2026-now

Teaching coding and informatics fundamentals in a part-time remote role, with personalized guidance for learners from diverse ages and backgrounds.

- Designed adaptive learning paths based on each learner's goals, pace, and prior experience.
- Applied project-based learning to guide learners in building contextual real-world projects while strengthening programming fundamentals.
- Improved learners' problem-solving, computational thinking, and logical reasoning through practical exercises.
- Encouraged independent and collaborative exploration to prepare learners for self-directed study, automation tasks, and professional use cases.

PROJECT EXPERIENCE

Aplikasi Koperasi Syariah Terintegrasi

Team Project – Fullstack Developer

Laravel, Vue.js, JavaScript, PHP

- Developed an integrated Sharia cooperative application with a complex business domain
- Applied feature-based, end-to-end (fullstack) task distribution instead of separating frontend and backend responsibilities
- Implemented a multi-role system with centralized access control
- Handled business workflows including savings, murabahah-based installment financing, POS cash transactions, and inventory management
- Designed data structures and application flows to ensure consistency between financial and inventory modules

Productivity Application (Java Desktop)

Team Project – Lead Developer

Java, JavaFX, PostgreSQL, SQLite

- Designed a desktop application using a layered MVC architecture
- Applied multiple design patterns to maintain code modularity and scalability
- Implemented a PostgreSQL database with SQLite fallback stored in the default user directory across operating systems
- Built a cross-platform, installable application, including Java runtime configuration
- Developed comprehensive unit testing and structured input–output validation
- Designed UI/UX with a focus on readability, usage flow, and overall user experience

Block Shooter Game

Team Project – Lead Developer

C, Raylib

- Led the development of a simple 2D game using C and raylib
- Manually implemented low-level data structures
- Designed the game loop, object behavior, and game state persistence
- Managed user experience considerations and team collaboration throughout development

ACHIEVEMENTS

ACM/ICPC Regional Asia <i>Honourable Mention</i>	2025
ACM/ICPC Provincial – Indonesia National Contest <i>Honourable Mention; Qualified for the ACM/ICPC Asia Regional Contest</i>	2025
Olimpiade Sains Nasional bidang Komputer/Informatika <i>District Champion; Qualified for the Provincial Level</i>	2022

ORGANIZATIONAL EXPERIENCE

Junior Staff Member – Computer Science Student Association

2025-sekarang

Politeknik Negeri Bandung (*Bandung State Polytechnic*)

- Assisted in planning and executing academic knowledge-sharing programs across study programs within the Informatics Engineering Department
- Involved in managing bootcamp activities and competition training programs, including participant management and academic coordination
- Supported the preparation and execution of student competitions, both administratively and technically

Head of Programming Division – School Computer Club

2022-2023

SMAN 1 Subang (*Senior High School 1 Subang*)

- Coordinated programming activities and collaborative projects among club members
- Designed and conducted training programs ranging from basic to advanced programming
- Mentored members in project development and problem-solving skill improvement
- Presented project outcomes at school events to encourage collaboration and innovation